Counting the Horizon: Area Units, Entropy, Energy, and Frequency

Light-Meter Physics

Abstract

For a Schwarzschild black hole, the horizon area can be expressed as a dimensionless ratio to the Planck area, $n_{\mathrm{A}} \equiv A/\ell_P^2$. The Bekenstein–Hawking law gives $S_{\mathrm{BH}}/k_{\mathrm{B}} = n_{\mathrm{A}}/4$, while expressing the same entropy in bits gives $n_{\mathrm{bit}} = n_{\mathrm{A}}/(4\ln 2)$. Combining the Hawking temperature with the Schwarzschild mass relation yields the exact identities $Mc^2 = (n_{\mathrm{A}}/2)k_{\mathrm{B}}T_H = 2\ln 2\, n_{\mathrm{bit}}k_{\mathrm{B}}T_H$. Dividing by Planck's constant gives equivalent relations among energy-over-h scales. We distinguish the area ratio, entropy in natural-log units, and entropy in bits, and explain why the half-$k_{\mathrm{B}}T_H$ form resembles equipartition without establishing physical degrees of freedom on the horizon. The identities assume a four-dimensional, asymptotically flat, uncharged, nonrotating black hole and standard semiclassical thermodynamic formulas; the area ratio is not a literal count of horizon tiles or stored bits.

Keywords: black-hole entropy, Planck area, information units, Hawking temperature, Smarr relation, equipartition

1 Introduction

A Schwarzschild black hole's mass-energy, Hawking temperature, horizon area, and entropy are linked by a short chain of standard relations. The horizon area divided by the Planck area is a dimensionless number. The Bekenstein–Hawking law assigns one quarter of that number to the entropy in units of k_{B}. Converting that entropy to bits introduces a factor of $\ln 2$. Combining the result with the Hawking temperature rewrites the mass-energy as an area-count-weighted thermal energy.

The purpose of this paper is to keep three quantities distinct: the dimensionless area ratio n_{A}, the entropy $S_{\mathrm{BH}}/k_{\mathrm{B}}$ in natural-log units, and the entropy value expressed in bits n_{bit}. The area ratio is geometric bookkeeping. The bit value is an entropy-unit conversion. Neither, by itself, specifies microscopic horizon constituents or physical memory cells.

The scope is a four-dimensional, asymptotically flat Schwarzschild black hole: zero spin and zero charge. We use the standard Hawking temperature normalized at infinity and Bekenstein–Hawking area entropy [1, 2, 3]. Any equipartition interpretation is identified separately as a hypothesis or analogy rather than as a consequence of the algebra. For a rotating Kerr black hole, the Smarr relation instead reads $Mc^2 = 2T_H S_{\mathrm{BH}} + 2\Omega_H J$, so the area-count thermal term corresponds to $Mc^2 - 2\Omega_H J$, not the entire mass-energy. The later Kerr manuscript develops this distinction [3].

2 Horizon area in Planck units

For mass M, the Schwarzschild radius, horizon area, and Planck area are

$$r_s \equiv \frac{2GM}{c^2}, \qquad A = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}, \qquad \ell_P^2 \equiv \frac{\hbar G}{c^3}. \tag{1}$$

Define the dimensionless Planck-area ratio

$$n_{\mathrm{A}} \equiv \frac{A}{\ell_P^2} = \frac{16\pi G M^2}{\hbar c}. \tag{2}$$

The symbol n_{A} denotes area expressed in Planck-area units. It is not assumed to be an integer, and defining it does not tile the horizon with physical squares. A grid used to illustrate the ratio is only an area ruler.

3 One area, two entropy units

The Bekenstein–Hawking relation assigns entropy to the horizon area [1, 2]:

$$S_{\mathrm{BH}} = \frac{k_{\mathrm{B}} A}{4\ell_P^2}. \tag{3}$$

Using Eq. (2), the entropy in dimensionless natural-log units is

$$\frac{S_{\mathrm{BH}}}{k_{\mathrm{B}}} = \frac{n_{\mathrm{A}}}{4}. \tag{4}$$

The quantity $S_{\mathrm{BH}}/k_{\mathrm{B}}$ is commonly described as entropy in nats, corresponding to a logarithm with base e.

One bit corresponds to an entropy increment $k_{\mathrm{B}} \ln 2$. Therefore the same entropy, expressed in bits, is

$$n_{\mathrm{bit}} \equiv \frac{S_{\mathrm{BH}}}{k_{\mathrm{B}} \ln 2} = \frac{n_{\mathrm{A}}}{4 \ln 2} \simeq 0.360674\, n_{\mathrm{A}}. \tag{5}$$

Equivalently, one entropy bit corresponds to $4 \ln 2 \simeq 2.77259$ Planck-area units under this convention. This base conversion follows the information-unit convention associated with Shannon entropy [4]. It does not establish that the horizon contains individually addressable binary switches, nor does it provide a microscopic model for black-hole states.

The phrase "half the bits" can therefore be misleading if read literally. The half in the energy formula below multiplies $k_{\mathrm{B}} T_H$ per unit of n_{A}; it does not mean that half of the entropy bits are counted.

4 The area-count energy identity

The Hawking temperature for a Schwarzschild black hole, referred to time at infinity, is [2]

$$T_H = \frac{\hbar c^3}{8\pi G M k_{\mathrm{B}}}. \tag{6}$$

Multiplying Eq. (2) by $k_{\mathrm{B}} T_H / 2$ gives

$$\frac{1}{2} n_{\mathrm{A}} k_{\mathrm{B}} T_H = \frac{1}{2} \left(\frac{16\pi G M^2}{\hbar c} \right) \left(\frac{\hbar c^3}{8\pi G M} \right) \tag{7}$$

$$= Mc^2. \tag{8}$$

Since $n_{\mathrm{A}} = 4 S_{\mathrm{BH}} / k_{\mathrm{B}}$, this is equivalently

$$Mc^2 = 2 T_H S_{\mathrm{BH}}. \tag{9}$$

This is the Schwarzschild form of the Smarr relation [3]. The derivation shows that the "per-area-unit" expression is the same thermodynamic identity regrouped; the constants cancel exactly within the stated formulas.

Using Eq. (5) to replace the area ratio instead gives

$$Mc^2 = 2 \ln 2\, n_{\mathrm{bit}} k_{\mathrm{B}} T_H. \tag{10}$$

The coefficient changes because the count changes convention. Equations (8) and (10) describe the same energy and temperature; neither convention identifies actual horizon constituents.

5 First-law interpretation of the factor

The Schwarzschild first law along a quasistatic family is

$$\mathrm{d}E = T_H \, \mathrm{d}S_{\mathrm{BH}}, \qquad E = Mc^2. \tag{11}$$

The area law gives $S_{\mathrm{BH}} \propto M^2 \propto E^2$. If $S_{\mathrm{BH}} = CE^2$ along this family, then

$$\frac{\mathrm{d}S_{\mathrm{BH}}}{\mathrm{d}E} = \frac{2S_{\mathrm{BH}}}{E}. \tag{12}$$

The first law also gives $\mathrm{d}S_{\mathrm{BH}}/\mathrm{d}E = 1/T_H$. Equating the two slopes yields $E = 2T_H S_{\mathrm{BH}}$. Thus the factor two records the quadratic entropy scaling; it is not an independent energy source or a separate equipartition derivation. This first-law argument concerns changes along the Schwarzschild family and does not alone fix the absolute entropy normalization.

6 Frequency form

An energy can be expressed in frequency units by dividing by Planck's constant. Define the mass-energy frequency and the Hawking thermal-energy frequency by

$$\nu_C \equiv \frac{Mc^2}{h}, \qquad f_H \equiv \frac{k_{\mathrm{B}} T_H}{h}, \qquad h = 2\pi\hbar. \tag{13}$$

Dividing Eq. (8) by h gives

$$\nu_C = \frac{1}{2} n_{\mathrm{A}} f_H = 2 \ln 2 \, n_{\mathrm{bit}} f_H. \tag{14}$$

Here ν_C is the Compton frequency corresponding to the mass-energy under the $E = h\nu$ convention; its angular-frequency counterpart is $Mc^2/\hbar = 2\pi\nu_C$. Likewise, f_H expresses the thermal energy in hertz. It is not the peak frequency of the Hawking spectrum, nor does Eq. (14) imply a physical oscillation of the horizon.

7 Equipartition form and its limits

Classical equipartition assigns an average energy $k_{\mathrm{B}}T/2$ to each independent quadratic degree of freedom in thermal equilibrium. Equation (8) has that algebraic form if n_{A} is treated as the number of such degrees of freedom at temperature T_H. However, no step in the derivation demonstrates that Planck-area units are independent quadratic modes, or that the horizon is

a classical material system in thermal equilibrium. That microscopic interpretation would require additional physical assumptions.

Horizon-equipartition interpretations have been proposed in broader thermodynamic or emergent-gravity frameworks [5, 6]. This manuscript uses them only as context. Its result is narrower: given the Schwarzschild area count, Hawking temperature, and Bekenstein–Hawking entropy, Eq. (8) follows algebraically. The equality does not determine the microstates or validate a particular microscopic theory.

8 Numerical examples

For illustration, use $GM_\odot = 1.3271244 \times 10^{20}\ \mathrm{m^3\,s^{-2}}$ and $GM_\oplus = 3.98600435507 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$ [7], together with $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ [8]. The values of c, h, and k_B are exact in the SI. Since G is measured, quantities involving the mass or Planck area are rounded. Both examples describe hypothetical Schwarzschild black holes with the Sun's and Earth's masses, not the actual astronomical bodies.

Table 1: Area, entropy, and energy-frequency quantities for hypothetical Schwarzschild black holes.

Quantity	Solar-mass black hole	Earth-mass black hole
Area ratio n_A	4.196×10^{77}	3.785×10^{66}
Entropy $S_\mathrm{BH}/k_\mathrm{B}$ (nats)	1.049×10^{77}	9.462×10^{65}
Entropy n_bit (bits)	1.513×10^{77}	1.365×10^{66}
Hawking temperature T_H	61.70 nK	20.54 mK
Thermal-energy scale f_H	1.286 kHz	428.1 MHz
Mass-energy $Mc^2 = 2T_H S_\mathrm{BH}$	1.787×10^{47} J	5.368×10^{41} J
Compton frequency ν_C	2.697×10^{80} Hz	8.101×10^{74} Hz

The solar-to-Earth mass ratio is about 3.329×10^5. The area ratio and entropy therefore scale by its square, about 1.109×10^{11}, while the Hawking temperature and thermal-energy frequency scale inversely with mass. These examples check the algebra at widely separated scales; they do not imply that either body's actual horizon has been measured or that the displayed frequencies are emitted spectral peaks.

9 Interpretation and limits

Three count-like quantities appear, but they answer different questions. The area ratio n_A expresses geometric area in Planck units. The dimensionless entropy $S_\mathrm{BH}/k_\mathrm{B}$ expresses the

same area law in natural-log units. The bit value $n_{\rm bit}$ expresses that entropy using logarithms to base two. The fixed ratios among them are consequences of definitions and the area law, not separate measurements of horizon structure.

The energy relation is specific to the Schwarzschild assumptions and to standard semi-classical black-hole thermodynamics. Rotation, charge, different spacetime dimensions, or different asymptotic conditions require their corresponding mass, temperature, and entropy relations. No claim is made here that the area units are physical tiles, that each unit stores a bit, or that the equipartition form reveals the microscopic degrees of freedom.

10 Conclusion

For a Schwarzschild horizon, $n_{\rm A} = A/\ell_P^2$ is a useful dimensionless area ratio. The Bekenstein–Hawking law converts it to entropy in natural-log units, $S_{\rm BH}/k_{\rm B} = n_{\rm A}/4$, or to entropy in bits, $n_{\rm bit} = n_{\rm A}/(4\ln 2)$. Together with the Hawking temperature, these definitions rewrite the mass-energy as $Mc^2 = (n_{\rm A}/2)k_{\rm B}T_H = 2\ln 2\, n_{\rm bit}k_{\rm B}T_H$.

The half-$k_{\rm B}T_H$ form resembles equipartition, but it is an algebraic restatement of the Schwarzschild Smarr relation. It does not prove that Planck-area units are independent degrees of freedom or literal information bits. Keeping area ratios, entropy units, and microscopic interpretations distinct makes the counting useful without overstating what it establishes.

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